HACKSHIFT HACKATHON

NexBridge.AI

Connecting Talent, Ideas, and Opportunities with AI

TEAM NAME : CODE CATALYST

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CODE CATALYST

THE PROBLEM

An intelligent opportunity ecosystem that connects talent, startups, mentors, and recruiters

FRAGMENTED OPPORTUNITY DISCOVERY

- Students and job seekers search across multiple platforms for jobs, internships, mentorships, and startup opportunities.
- No unified platform connects talent, skills, ideas, and opportunities in one ecosystem.

STARTUP IDEATION & VALIDATION GAP

- Aspiring entrepreneurs struggle to validate ideas, assess market demand, and identify competitors before investing time and resources.
- Users lack AI-powered guidance to refine ideas, build business models, form teams, and turn concepts into viable ventures.

DIFFICULTY IN STARTUP TEAM FORMATION

- Entrepreneurs and startups struggle to find collaborators with complementary technical and business skills.
- Students and aspiring founders lack an effective way to connect with mentors and potential co-founders.

SKILL–OPPORTUNITY MISMATCH

- Existing platforms primarily focus on individual jobs or professional connections rather than interconnected opportunities.
- Users lack AI-powered guidance to build skills, discover relevant connections, and turn ideas into viable ventures.

KEY DATA

ONLY 42.6%

Indian Graduates Are Considered Employable

The World Economic Forum's Future of Jobs Report 2025 found that 63% of surveyed employers identified skills gaps as a major barrier to business transformation between 2025 and 2030. World Economic Forum
Source: World Economic Forum — Future of Jobs Report 2025

63%

Employers Identify Skills Gaps as a Major Barrier

The World Economic Forum's Future of Jobs Report 2025 found that 63% of surveyed employers identified skills gaps as a major barrier to business transformation between 2025 and 2030. World Economic Forum
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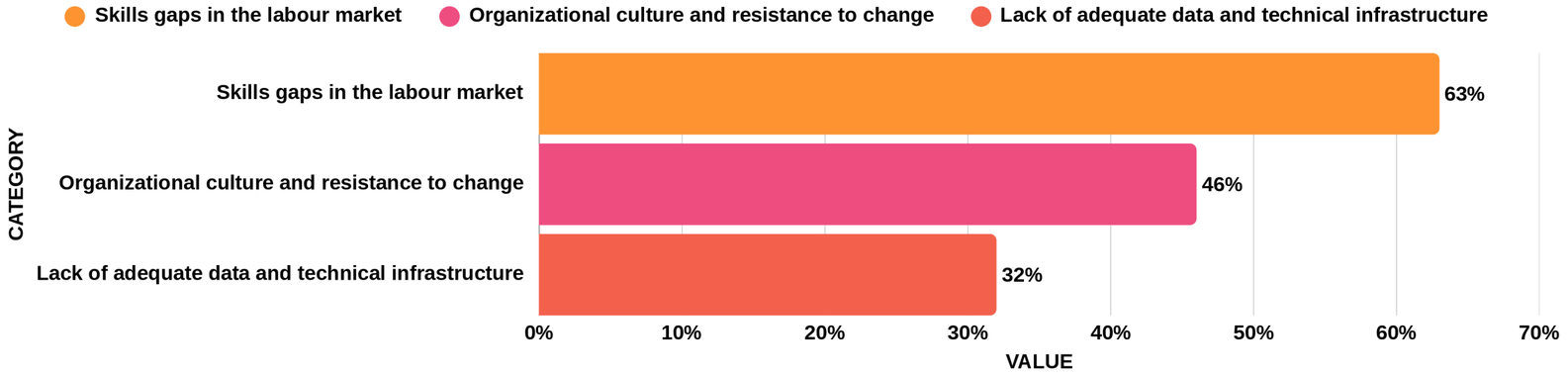
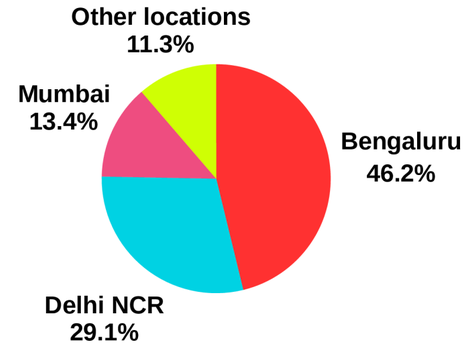


What Prevents Businesses from Unlocking Talent & Growth?

World Economic Forum, Future of Jobs Report 2025



India's Startup Funding Is Concentrated in 3 Major Hubs

Share of total startup funding by location, 2014–June 2024.

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OUR SOLUTION

Every Campus Event, One Intelligent Timeline.

NexBridge.AI is an AI-powered platform that connects students, job seekers, entrepreneurs, startups, mentors, and recruiters through intelligent opportunity matching. It helps users discover relevant jobs, develop in-demand skills, and build complementary startup teams. By connecting talent, ideas, and opportunities in one ecosystem, VentureOS AI turns potential into real-world impact.

AI-POWERED TALENT INTELLIGENCE

- Creates personalized profiles based on users' skills, experience, projects, and career goals.
- Identifies skill gaps and recommends relevant opportunities to help users grow professionally.

IDEA-TO-INCUBATION ENGINE

- AI-powered idea evaluation, refinement, and validation transforms raw concepts into structured business proposals.
- End-to-end venture guidance supports market research, business model development, team formation, pitch preparation, and incubator connections.

AI STARTUP TEAM FORMATION

- Analyzes startup ideas to identify the key roles and skills required to build the venture.
- Connects founders with potential co-founders and collaborators who have complementary skills.

PERSONALIZED SKILL DEVELOPMENT

- Compares users' existing skills with the requirements of their target opportunities.
- Recommends relevant projects, learning paths, and practical experiences to improve employability.

SMART MENTOR & RECRUITER MATCHING

- Connects aspiring entrepreneurs and job seekers with relevant mentors, recruiters, and industry professionals.
- Recommends connections based on expertise, career interests, industry, and professional goals.

UNIFIED OPPORTUNITY RADAR

- Brings jobs, internships, startup competitions, hackathons, grants, and incubator programs into one discovery hub.
- Highlights relevant opportunities, application deadlines, eligibility criteria, and important requirements.

AI-POWERED CAREER GUIDANCE

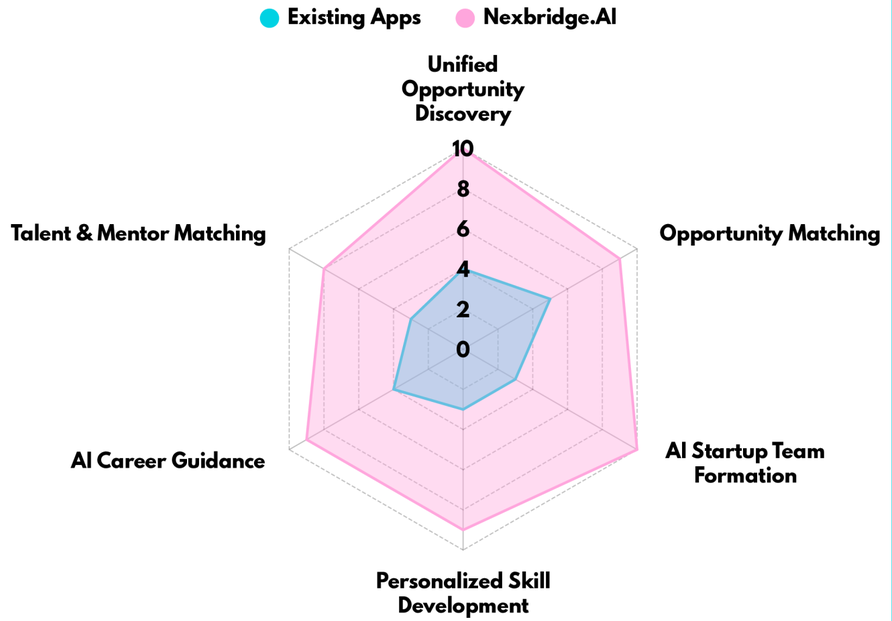
- Helps users explore career paths based on their current skills, interests, and long-term ambitions.
- Suggests actionable next steps to bridge the gap between their current profile and desired opportunities.

OPPORTUNITY GRAPH AI

- Builds intelligent connections between people, skills, projects, mentors, startups, and employment opportunities.
- Reveals personalized pathways showing how users can turn existing skills and ideas into new entrepreneurial opportunities.

PERSONALIZED OPPORTUNITY DASHBOARD

- Provides a centralized view of recommended opportunities, saved listings, applications, and professional progress.
- Prioritizes relevant matches and helps users track their journey from discovery to application or venture creation.



Platform Capability Comparison

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FLOW OF SOLUTION

STAGE 1: USER INPUT
Accepts student/job-seeker for profiles, skills, experience, career goals, portfolios, mentor/recruitor requirements.

STRUCTURED USER INFORMATION

STAGE 2: PROFILE INTELLIGENCE
Input Validation
Profile Structuring
Skill Extraction
Experience Normalization
Goal Identification

STRUCTURED TALENT PROFILE

STAGE 10: EXTERNAL OPPORTUNITY SOURCES
Feeds job, internship, startup/project, hackathon, and mentor/incubator programs into Stage 3.

STAGE 3: OPPORTUNITY RETRIEVAL LAYER
Search & retrieve, extract info, filter/rank, and validate eligibility.

NORMALIZED PROFILES & OPPORTUNITIES

STAGE 4: DATA PREPARATION & NORMALIZATION
Consolidates Outputs from Stages 2 & 3
duplicate detection, taxonomy normalization, metadata validation, and standardizing profiles & opportunities.

STAGE 5: PARALLEL AI INTELLIGENCE MODULES

- **MODULE A: TALENT INTELLIGENCE** — Extracts skills, experience, and strengths.
- **MODULE B: SEMANTIC OPPORTUNITY MATCHING** — Matches skills to requirements.
- **MODULE C: SKILL-GAP ANALYSIS** — Identifies development priorities.
- **MODULE D: STARTUP TEAM FORMATION** — Matches complementary co-founders.
- **MODULE E: MENTOR & RECRUITER MATCHING** — Aligns expertise and goals.
- **MODULE F: CAREER & VENTURE GUIDANCE** — Suggests paths and steps.

STAGE 6: MATCHING & VALIDATION ENGINE
Weighted compatibility scoring, requirements comparison, interest/goal alignment, match ranking.

RANKED & EXPLAINABLE MATCHES

STAGE 7: PERSONALIZED RECOMMENDATION ENGINE
Personalized opportunity ranking, career/venture recommendations, skill suggestions.

PERSONALIZED OPPORTUNITY RESULTS

STAGE 12: PERSONALIZED OPPORTUNITY ECOSYSTEM
Offers 2 pathways:

CAREER JOURNEY
Skills → Opportunities → Development Employment

&

VENTURE JOURNEY
Idea → Refinement → Team Formation → MVP Roadmap → Incubator

Bidirectional mapping for teams/mentors →

← Retrieval for startups

STAGE 13: IDEA-TO-INCUBATION ENGINE (Entrepreneurial Workflow)

1. **RAW IDEA INPUT** — Founder describes initial business idea
2. **AI IDEA REFINEMENT** — Clarifies problem, users & solution
3. **MARKET & COMPETITOR RESEARCH** — Identifies competitors & market opportunities
4. **IDEA VALIDATION** — Evaluates feasibility, needs, risks & experiments
5. **BUSINESS MODEL DESIGN** — Develops value prop, segments, revenues
6. **TEAM FORMATION** — Identifies roles & matches potential co-founders via Talent Intelligence
7. **MVP PLANNING** — Defines features, tech & roadmap
8. **PITCH DECK GENERATION** — Problem, solution, market & product roadmap
9. **INCUBATOR & MENTOR MATCHING** — Recommends incubators, accelerators & funding

FINAL OUTPUT: INCUBATION-READY VENTURE PROFILE
A guided pathway, not an automatic guarantee of market validation, funding, or incubation acceptance

INPUT → PROCESSING → AI INTELLIGENCE → MATCHING → PERSONALIZATION → OPPORTUNITIES

TECH STACK & ARCHITECTURE

Nexbridge.AI combines LLMs, semantic embeddings, intelligent matching algorithms, and a centralized database to power personalized opportunity discovery, skill-gap analysis, and AI-driven startup team formation—transforming fragmented talent data into actionable insights through a scalable, API-driven architecture.

AI & INTELLIGENCE ENGINE

- **LLM Integration:** Generates profile insights, explains opportunity matches, and powers AI career guidance.
- **Semantic Matching:** Uses embeddings to match users with jobs, mentors, projects, and potential co-founders.
- **Skill Intelligence:** Extracts skills, identifies skill gaps, and recommends personalized learning paths.

TALENT & OPPORTUNITY INTELLIGENCE

- **Profile Processing:** Structures user skills, experience, interests, projects, and career goals.
- **Matching Algorithm:** Calculates compatibility scores using skills, interests, experience, and opportunity requirements.
- **Team Formation:** Identifies complementary skill sets and recommends potential startup collaborators.

BACKEND & PROCESSING

- **FastAPI (Python):** Provides REST APIs for AI services, opportunity matching, and profile management.
- **Data Processing:** Handles skill extraction, profile analysis, opportunity filtering, and match ranking.
- **API Integration:** Connects the frontend, AI models, database, and external opportunity sources.

DATABASE & STORAGE

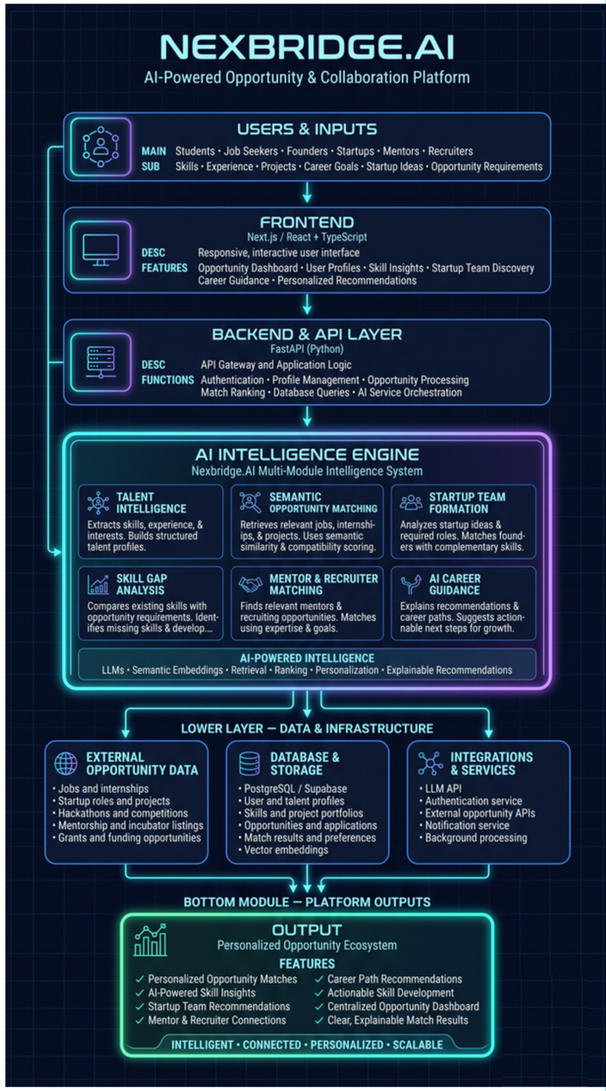
- **PostgreSQL / Supabase:** Stores user profiles, skills, opportunities, applications, and mentor information.
- **Authentication:** Supports secure sign-in, user sessions, and access control.
- **Vector Storage:** Stores semantic embeddings to support similarity search and intelligent matching.

FRONTEND & USER EXPERIENCE

- **React / Next.js + TypeScript:** Builds a responsive, component-based web application.
- **Tailwind CSS:** Provides consistent styling, reusable components, and responsive layouts.
- **Interactive Dashboard:** Displays personalized matches, opportunity listings, skill gaps, and team recommendations.

SYSTEM ARCHITECTURE

- **Modular Architecture:** Separates frontend, backend APIs, AI services, and database layers.
- **Intelligent Workflow:** Connects profile analysis, semantic retrieval, compatibility scoring, and personalized recommendations.
- **Deployment & Scalability:** Supports cloud deployment, API-based integrations, and future feature expansion.



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INNOVATION & IMPACT

FEASIBILITY

- **Technical Feasibility:** Built using React/Next.js, FastAPI, PostgreSQL/Supabase, and LLM APIs to power intelligent opportunity matching and career guidance.
- **AI-Driven Intelligence:** Semantic matching, skill-gap analysis, personalized recommendations, and AI-powered startup team formation connect talent with relevant opportunities.
- **Data Accessibility:** Structured user profiles and opportunity listings enable efficient matching across jobs, internships, projects, mentorships, and startup roles.
- **Scalable Modular Architecture:** Independent frontend, backend, database, and AI components support future expansion and third-party integrations.
- **Hackathon MVP Feasibility:** Core features can be demonstrated within 24 hours using a focused scope, sample opportunity data, and a working end-to-end workflow.
- **Real-World Deployment Potential:** Cloud hosting, secure authentication, and extensible APIs provide a foundation for future production deployment.

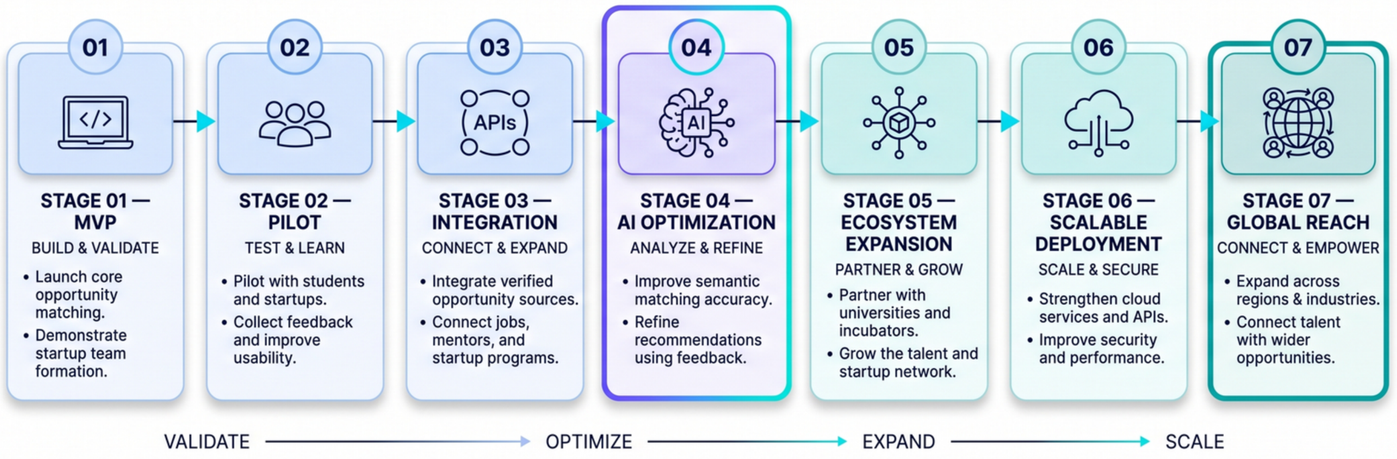
NEXBRIDGE.AI

WHAT MAKES US DIFFERENT?

One Intelligent Ecosystem for Talent, Careers, and Startups

CAPABILITY	NEXBRIDGE.AI	JOB PORTALS	PROFESSIONAL NETWORKS	GENERAL AI CHATBOTS
Intelligent Opportunity Matching Matches people with relevant opportunities using their profiles.	✓	✓	⚠	⚠
Skill-Gap Analysis Identifies skills to develop for target roles & opportunities.	✓	⚠	⚠	⚠
AI Startup Team Formation Connects founders with collaborators having complementary skills.	✓	✗	⚠	⚠
Personalized Career Guidance Suggests career paths and actionable next steps.	✓	⚠	⚠	✓
Unified Opportunity Discovery Brings jobs, internships, projects, and startup opportunities together.	✓	⚠	⚠	⚠
Mentor & Recruiter Matching Recommends relevant professional connections.	✓	⚠	✓	⚠
Semantic Skill Matching Connects people and opportunities based on skill meaning and context.	✓	⚠	⚠	⚠
Personalized Skill Development Recommends learning paths and practical projects.	✓	⚠	⚠	✓
Explainable Match Recommendations Explains why a particular opportunity may suit a user.	✓	⚠	⚠	⚠
Talent-to-Venture Connections Helps connect talent, startup ideas, and potential founding teams.	✓	✗	⚠	⚠

SCALE-UP FLOW



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BUSINESS MODEL

BUSINESS MODEL

SUBSCRIPTION FOR ACADEMIES
- Institutional licensing for campus-wide access to talent and opportunity matching.
- Premium analytics for career services and e-cells.

FREEMIUM USER ACCESS
- Free profiles, opportunity discovery, and basic AI-powered matching.
- Premium career guidance, advanced skill-gap analysis, and personalized recommendations.

RECRUITER & STARTUP PLANS
- Paid access to targeted talent discovery and candidate matching.
- Premium tools for startup team formation and skill-based recruitment.

PARTNERSHIP REVENUE
- Collaborations with universities, incubators, accelerators, and startup communities.
- Sponsored opportunities and recruitment partnerships

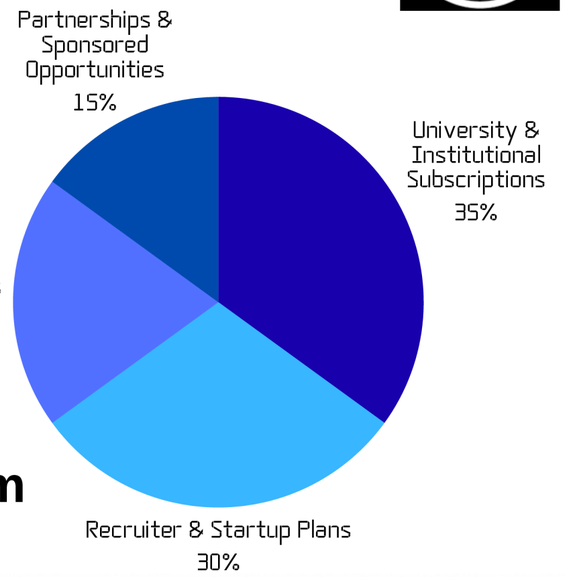
PRACTICAL APPLICATIONS IN NexBridge.AI

INTELLIGENT TALENT MATCHING
- AI-powered matching between skills and relevant opportunities
- Personalized job, internship, and project recommendations

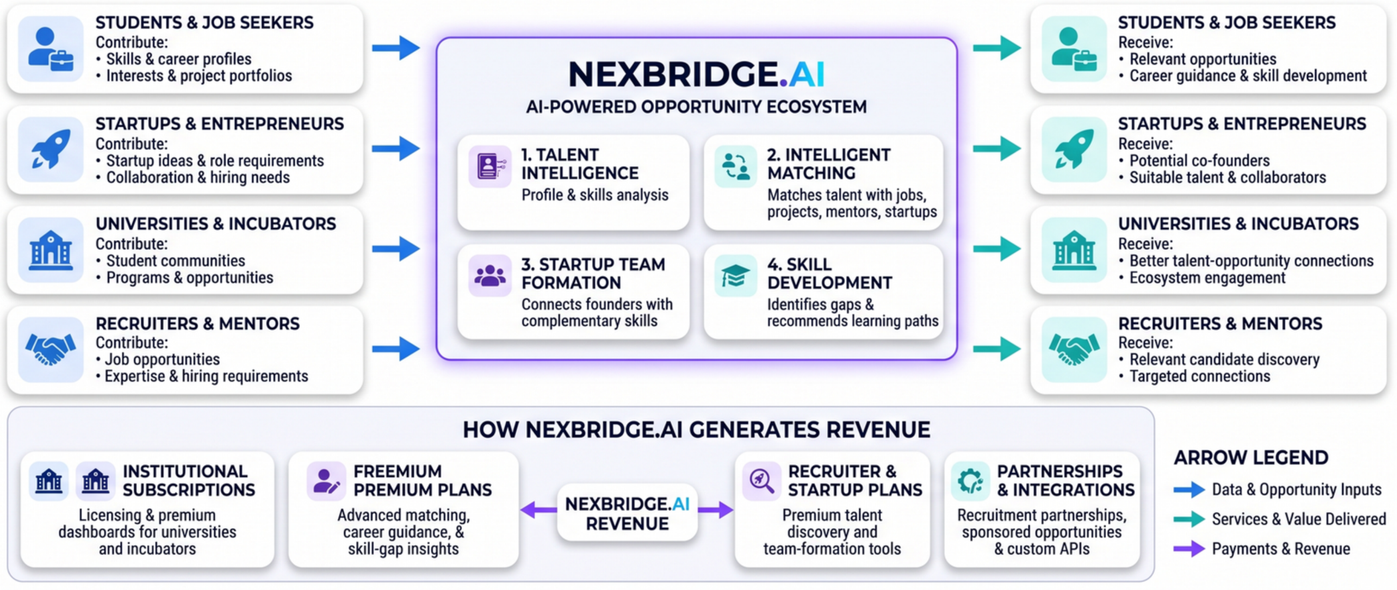
STARTUP TEAM FORMATION
- Connect founders with complementary talent and potential co-founders
- Match startup ideas with people who have relevant skills

CAREER & ECOSYSTEM GROWTH
- Identify skill gaps and recommend personalized learning paths
- Connect students, recruiters, mentors, and startups in one ecosystem

Revenue Streams



Value Flow Diagram



SCALE-UP FLOW

THANK YOU